Supplementary Information

Sharp Target-Domain Certificates for Quantum-Kernel Advantage under Distribution Shift

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Supplementary Results: Scope

This document contains extended results, complete sensitivity tables, and diagnostics supporting the main article. No procedure is defined exclusively here: all data construction, preprocessing, candidate grids, selection rules, inference, external acquisition, geometry calculations, and finite-shot procedures are reported in the main manuscript Methods. Machine-readable values and frozen input hashes are included in the immutable release artifact.

Supplementary Results: Target-label evidence frontier

The main text reports the 0.010 materiality threshold. Supplementary Table 1 gives the fixed adaptive audit at 0.005 and 0.020; Supplementary Table 2 compares strong prediction-aware controls, the exact ex-post oracle, and the weak prediction-independent reference at 0.010. A value of zero means the zero-label certificate already crosses the threshold. Every adaptive count is deterministic for the fixed prediction matrix, labels, acquisition rule, and tie order. Comparator medians summarize 200 prespecified SHA-256 tie orders and are algorithmic sensitivities, not confidence intervals.

The prespecified reference-breadth sensitivity uses complete five-dimension kernel blocks and 5,000 SHA-256-derived nested block orderings. It therefore contains three evaluated breadths rather than only the customary and full endpoints. The final Supplementary table summarizes the median zero-label endpoint within each case–classifier cell over those orderings and then across the 16 cells. Candidate count is a measure of reference breadth only, not time, memory, or computational cost.

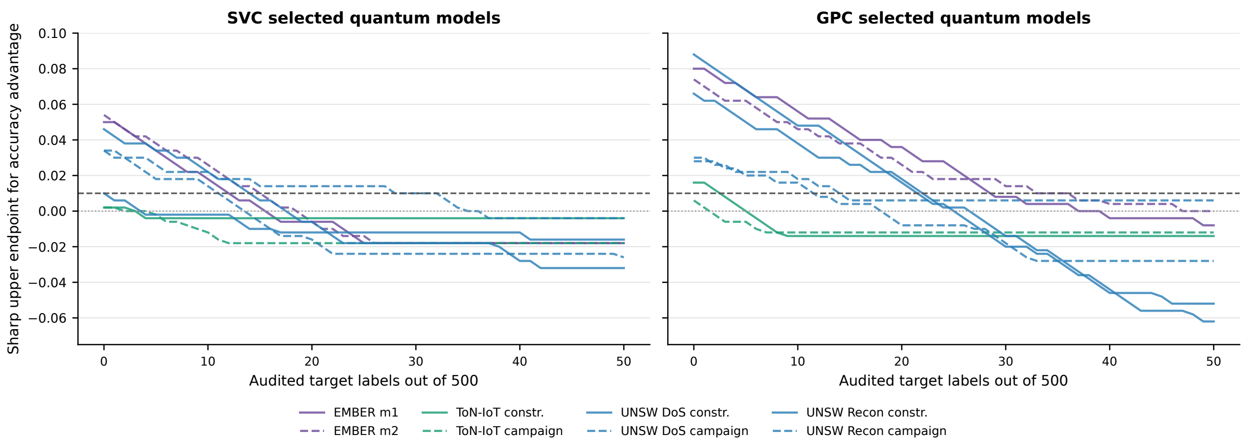
Supplementary Table 1: Target labels required against the full 115-candidate classical family. Columns $n_{.005}$, $n_{.010}$, and $n_{.020}$ use the fixed adaptive coverage audit. Every target batch contains 500 points.

Target case	Classifier	$n_{.005}$	$n_{.010}$	$n_{.020}$
EMBER $m1$	SVC	15	12	10
EMBER $m2$	SVC	17	15	12
ToN-IoT constructed	SVC	0	0	0
ToN-IoT campaign	SVC	0	0	0
UNSW DoS constructed	SVC	3	0	0
UNSW DoS campaign	SVC	13	11	5
UNSW Recon constructed	SVC	17	14	11
UNSW Recon campaign	SVC	34	28	11
EMBER $m1$	GPC	32	29	26
EMBER $m2$	GPC	40	33	23
ToN-IoT constructed	GPC	4	3	0
ToN-IoT campaign	GPC	1	0	0
UNSW DoS constructed	GPC	24	22	20
UNSW DoS campaign	GPC	14	12	5
UNSW Recon constructed	GPC	23	22	19
UNSW Recon campaign	GPC	54	14	10

Supplementary Table 2: Acquisition controls at $U \le 0.010$. Oracle is the exact retrospective minimum; adaptive is the fixed bottleneck-coverage rule; random active and initial coverage are medians across 200 SHA-256 tie orders; hash all is the weak prediction-independent median. The adaptive rule does not dominate either strong control.

Target case	Classifier	Oracle	Adaptive	Random active	Initial coverage	Hash all
EMBER $m1$	SVC	10	12	14.0	14.0	267.0
EMBER $m2$	SVC	11	15	16.0	16.0	284.0
ToN-IoT constructed	SVC	0	0	0.0	0.0	0.0
ToN-IoT campaign	SVC	0	0	0.0	0.0	0.0
UNSW DoS constructed	SVC	0	0	0.0	0.0	0.0
UNSW DoS campaign	SVC	6	11	8.0	8.0	220.0
UNSW Recon constructed	SVC	9	14	13.0	13.0	234.0
UNSW Recon campaign	SVC	7	28	28.0	286.5	312.5
EMBER $m1$	GPC	18	29	34.0	34.0	369.5
EMBER $m2$	GPC	16	33	34.0	34.0	371.5
ToN-IoT constructed	GPC	2	3	2.0	2.0	112.0
ToN-IoT campaign	GPC	0	0	0.0	0.0	0.0
UNSW DoS constructed	GPC	14	22	21.0	21.0	244.5
UNSW DoS campaign	GPC	5	12	11.0	11.0	246.5
UNSW Recon constructed	GPC	20	22	25.0	24.0	260.0
UNSW Recon campaign	GPC	5	14	12.0	12.0	352.5
Median over cells		6.5	13.0	12.5	12.5	245.5

The frozen initial-coverage policy has one severe failure: for UNSW Recon campaign SVC, its median is 286.5 labels because the zero-label witness set becomes stale. Recomputing the actionable witness set avoids that failure, but optimizing tied-witness coverage gives no consistent advantage over dynamic random-active sampling.



Supplementary Figure 1: Witness-directed contraction of the sharp accuracy upper endpoint against the full classical family. Each curve is one fixed target batch and selected quantum model; colour denotes source dataset, and solid/dashed lines separate its paired shifts. Labels audit the fixed comparison and never retune or select a model. Curves can cross below zero because the quantum classifier can become certifiably worse than the best classical witness. At 500 labels every curve equals the realized advantage.

Supplementary Table 3: Canonical fixed runs and P1′ quantum configurations used by every target certificate. The exact run identifier is the displayed prefix followed by __ms42__q1000_id500_ood500__qs42__s42; a slash in a prefix denotes a literal __ separator. Configuration columns identify the classifier and report map, scale, dimension, and the selected SVC C where applicable.

Target case	Run prefix	SVC configuration	GPC configuration
EMBER $m1$	m1_hist_byteent	pauli_xz_r1_full/as2/d12/C10	zz_r1_full/as0.5/d12
EMBER $m2$	m2_hist_byteent	pauli_xz_r1_full/as2/d12/C100	zz_r2_full/as2/d8
ToN-IoT constructed	toniot_scanning/m2_centroid	zz_r2_full/as2/d6/C10	zz_r1_full/as2/d6
ToN-IoT campaign	toniot_scanning/natural_cur	pauli_xz_r1_full/as2/d12/C10	zz_r2_full/as0.5/d10
UNSW DoS constructed	unsw_dos/m2_centroid	zmap_r2/as0.5/d12/C100	zz_r1_full/d12
UNSW DoS campaign	unsw_dos/natural_cur	zz_r2_full/as0.5/d4/C100	zz_r1_full/as0.5/d4
UNSW Recon constructed	unsw_recon/m2_centroid	zz_r2_full/as0.5/d4/C100	zz_r2_full/as0.5/d8
UNSW Recon campaign	unsw_recon/natural_cur	zz_r1_full/as0.5/d6/C10	zz_r2_full/as0.5/d4

Supplementary Results: Prevalence-conditional balanced accuracy

Supplementary Table 4 reports the exact finite-batch MILP after fixing prevalence to the target value that becomes available when the label file is unlocked. It is a retrospective metric sensitivity, not a label-free use of the true prevalence. The LP relaxation and integer program agree to numerical precision in all reported cells. Because these batches are close to balanced by construction, the balanced-accuracy upper endpoints closely track their accuracy counterparts and do not rescue a broad zero-label certificate.

Supplementary Table 4: Prevalence-conditional sharp balanced-accuracy upper endpoints against the full classical family. U_{Acc} is the assumption-free zero-label accuracy endpoint, $U_{\mathrm{BAcc}}(\pi)$ is the exact MILP endpoint at the retrospectively unlocked prevalence, and $\Delta_{\mathrm{BAcc,full}}$ is the realized balanced-accuracy advantage.

Target case	Classifier	π	U_{Acc}	$U_{\mathrm{BAcc}}(\pi)$	$\Delta_{\mathrm{BAcc,full}}$
EMBER $m1$	SVC	.488	.050	.050	−.024
EMBER $m2$	SVC	.488	.054	.054	−.037
ToN-IoT constructed	SVC	.492	.002	.002	−.004
ToN-IoT campaign	SVC	.518	.002	.002	−.018
UNSW DoS constructed	SVC	.496	.010	.010	−.016
UNSW DoS campaign	SVC	.520	.034	.035	−.025
UNSW Recon constructed	SVC	.498	.046	.046	−.032
UNSW Recon campaign	SVC	.470	.034	.033	−.022
EMBER $m1$	GPC	.488	.080	.078	−.033
EMBER $m2$	GPC	.488	.074	.073	−.016
ToN-IoT constructed	GPC	.492	.016	.016	−.014
ToN-IoT campaign	GPC	.518	.006	.006	−.012
UNSW DoS constructed	GPC	.496	.066	.066	−.052
UNSW DoS campaign	GPC	.520	.028	.029	−.033
UNSW Recon constructed	GPC	.498	.088	.088	−.066
UNSW Recon campaign	GPC	.470	.030	.029	+.001

Supplementary Results: Prospective Gate-2 replication

The Gate-2 protocol was frozen before acquiring any of its three tasks, computing a model prediction, or opening a target label. Its specification SHA-256 is 3a8318d92d4af2aeeaf0c0edb069c3be 59f31da6d1ee50fb6a6256e9d9d280b0. The prediction stage copied target labels into a separate sealed tree and then removed them from the fitting state. Once every available unit contained 175 candidate checkpoints for both classifiers, the aggregate prediction lock was written with SHA-256 e4ee6bb89bc78047e7c848a0db3a6255cf0a2600a21b42fbab956a0e889507d1. Only then did the audit process record its one-time label opening. The tasks were inherited unchanged from the unused fallback list in the earlier external protocol.

During acquisition, the pinned NHANES download URLs returned HTML 404 pages. The transport paths alone were updated to the current official CDC directory; the same 37 cycle-specific XPT filenames, TableShift parser, columns, and domain definition were retained. XPORT magic bytes and SHA-256 hashes were verified before export, and the complete URL-repair audit is archived. This maintenance preceded every model execution and did not alter the subsequent minimum-feature disposition.

Supplementary Table 5: Technical disposition of every frozen prospective Gate-2 task. "Candidates" counts kernel configurations per available seed; each configuration supplies SVC and GPC predictions. NHANES was retained as a recorded technical failure and was not replaced.

Task	Shift	Status	Seeds	Candidates/seed
BRFSS Diabetes	race	prediction-locked	5	175
ACS Food Stamps	Census division	prediction-locked	5	175
NHANES Lead	poverty	minimum-feature failure ($11 < 12$)	5	0

Supplementary Table 6 gives every seed-level primary cell. The sharp endpoint is already at most 0.010 without labels in 11/20 cells. Across all cells, the adaptive audit has median zero and range 0–76; the exact label oracle has median zero and range 0–40. Every realized accuracy effect is negative. The one broad zero-label case, ACS Food Stamps SVC at seed 999, is retained and determines the maximum adaptive count.

Supplementary Table 6: Every prospective full-family cell at the primary accuracy threshold. U_0 is the zero-label sharp upper endpoint, $n_{.01}$ is the fixed adaptive count, $n^*_{.01}$ is the exact retrospective label-oracle minimum, and Δ_{full} is the realized quantum-minus-best-classical accuracy difference after label opening.

Task	Classifier	Seed	U_0	$n_{.01}$	$n^*_{.01}$	Δ_{full}
BRFSS Diabetes	SVC	42	.012	3	1	−.154
	SVC	123	.008	0	0	−.156
	SVC	999	.008	0	0	−.110
	SVC	7	.024	12	4	−.096
	SVC	2024	.018	4	2	−.118
	GPC	42	.000	0	0	−.008
	GPC	123	.002	0	0	−.010
	GPC	999	.004	0	0	−.018
	GPC	7	.002	0	0	−.002
	GPC	2024	.018	2	2	−.026
ACS Food Stamps	SVC	42	.010	0	0	−.030
	SVC	123	.014	1	1	−.060
	SVC	999	.168	76	40	−.104
	SVC	7	.010	0	0	−.108
	SVC	2024	.006	0	0	−.076
	GPC	42	.010	0	0	−.004
	GPC	123	.006	0	0	−.018
	GPC	999	.022	3	3	−.014
	GPC	7	.032	10	6	−.010
	GPC	2024	.024	4	4	−.006

The frozen outcome summaries are the medians over five seeds in each task–classifier cell. They are 3, 0, 0, and 3 labels for BRFSS SVC, BRFSS GPC, ACS SVC, and ACS GPC, respectively. All four are at most 50, the overall seed-level median is zero, and the maximum is 76, satisfying every numerical condition for strong prospective transfer. The reported category remains `strong prospective transfer---technically limited two-task replication` because only two tasks completed the grid.

Supplementary Table 7: Prospective acquisition controls at $U \leq 0.010$. Adaptive entries are median [minimum, maximum] over five seeds. Other entries are median [2.5th, 97.5th percentile] over 1,000 seed–order combinations. Ranges are algorithmic sensitivities, not confidence intervals.

Task	Classifier	Adaptive	Random active	Initial coverage	Hash all
BRFSS Diabetes	SVC	$3[0, 12]$	$2[0, 12]$	$2[0, 12]$	$91[0, 301]$
	GPC	$0[0, 2]$	$0[0, 2]$	$0[0, 2]$	$0[0, 162]$
ACS Food Stamps	SVC	$0[0, 76]$	$0[0, 79]$	$0[0, 78]$	$0[0, 350]$
	GPC	$3[0, 10]$	$4[0, 15]$	$4[0, 15]$	$141[0, 415]$

Adaptive coverage wins, ties, and loses against each strong comparator in 3, 15, and 2 seed-level cells; its median paired difference is zero. This prospectively repeats the conclusion that actionable-disagreement restriction is useful while coverage optimization is not demonstrably superior. The realized-prevalence balanced-accuracy MILP is retained as a retrospective sensitivity. Its full-family upper endpoints span 0–0.223, whereas every realized balanced-accuracy effect is negative (range -0.080 to -0.013). Because prevalence was not known before label opening, these values are not prospective label-free balanced-accuracy certificates.

All 20 selected prospective quantum models are product maps. Consequently, this experiment validates transfer of Gate 2 on the eligible fixed tasks but does not independently replicate the retrospective entangling-ZZ result.

Supplementary Results: Cross-generation sensitivity map

The ten rows remain in their prespecified S1–S10 order rather than being ranked by outcome. Every row includes all eight fixed security scenario-groups. Scenario-groups are averaged within EMBER, UNSW-NB15, and ToN-IoT before the three source datasets receive equal weight. The corrected S10 values in Table 8 are identical to the confirmatory endpoint in the main article. This display compares complete analysis specifications across two result generations; it is not a factorial decomposition of their individual choices.

Supplementary Table 8: Prespecified cross-generation definitions and source-dataset-equal quantum-minus-classical OOD balanced-accuracy effects. "Oracle" selects and evaluates on the same OOD test labels; "ID-val" uses a disjoint in-distribution validation split. SVC regularization is fixed at $C = 1$ or selected by training-only cross-validation; this axis is not applicable to GPC. S1–S4 use the legacy generation and S5–S10 use v4, so their boundary does not isolate the effect of regularization. Values are descriptive fixed-case summaries, not per-axis causal effects.

ID	Selection	SVC C	Classical reference	Budget	SVC	GPC
S1	OOD oracle	Fixed	Linear/RBF	Native	$+0.0313$	$+0.0340$
S2	Legacy ID	Fixed	Linear/RBF	Native	$+0.0475$	$+0.0418$
S3	OOD oracle	Fixed	Extended	Native	$+0.0058$	$+0.0145$
S4	Legacy ID	Fixed	Extended	Native	$+0.0014$	$+0.0122$
S5	OOD oracle	Train CV	Linear/RBF	Native	$+0.0034$	$+0.0125$
S6	ID-val	Train CV	Linear/RBF	Native	-0.0041	$+0.0059$
S7	OOD oracle	Train CV	Extended	Native	-0.0054	$+0.0012$
S8	ID-val	Train CV	Extended	Native	-0.0038	-0.0029
S9	OOD oracle	Train CV	Extended	Equal	-0.0032	$+0.0038$
S10	ID-val	Train CV	Extended	Equal	-0.0056	-0.0009

The source-dataset-equal range is 0.0531 for SVC and 0.0447 for GPC. This spread is not an interval: the specifications share data and candidates and deliberately target different estimands.

Supplementary Results: Candidate-budget sampling

Table 9 shows every group–classifier cell under all three prespecified sampling schemes. The primary blocked scheme mirrors the five-dimension block structure of the quantum pool; the other schemes match candidate count but not the same search-space structure. Their close values show that the controlled conclusion is not an artefact of the blocked draw.

Supplementary Table 9: Complete equal-budget sampling sensitivity. Entries are median quantum-minus-classical P1′ OOD differences over 5,000 classical-pool subsamples. “Blocked” is primary; “uniform” and “stratified” are sensitivities. Source-dataset-equal rows average scenario-group medians within the three source datasets. These resampling medians are distinct from the run-level expected-classical endpoint used for the conditional intervals in the main article.

Scenario	Clf.	Blocked	Uniform	Stratified
EMBER $m1$	SVC	−0.0056	−0.0054	−0.0054
	GPC	−0.0004	−0.0009	−0.0013
EMBER $m2$	SVC	−0.0071	−0.0071	−0.0071
	GPC	+0.0007	+0.0008	+0.0005
UNSW-DoS (campaign)	SVC	−0.0036	−0.0034	−0.0036
	GPC	−0.0072	−0.0072	−0.0071
UNSW-DoS (constructed)	SVC	−0.0046	−0.0030	−0.0028
	GPC	−0.0116	−0.0113	−0.0112
UNSW-Recon (campaign)	SVC	+0.0050	+0.0052	+0.0052
	GPC	+0.0046	+0.0047	+0.0049
UNSW-Recon (constructed)	SVC	−0.0090	−0.0089	−0.0089
	GPC	−0.0082	−0.0076	−0.0077
ToN-IoT (campaign)	SVC	−0.0147	−0.0149	−0.0143
	GPC	−0.0095	−0.0094	−0.0096
ToN-IoT (constructed)	SVC	−0.0011	−0.0008	−0.0011
	GPC	+0.0093	+0.0063	+0.0057
Source-dataset-equal	SVC	−0.00576	−0.00555	−0.00548
	GPC	−0.00185	−0.00232	−0.00253

The largest within-cell scheme range is 0.00353 for ToN-IoT constructed under GPC. The next largest is 0.00184 for UNSW-DoS constructed under SVC.

Supplementary Results: Effective-rank matching sensitivities

The predefined rank ratio is $\max(r_Q, r_C)/\min(r_Q, r_C)$. Table 10 reports all calipers for both observational matchers. With replacement, 75.2% of pairs survive the primary 1.25 caliper. One-to-one assignment has a smaller common support and becomes visibly poor without a caliper (95th-percentile ratio 11.45), making the overlap restriction scientifically material.

Supplementary Table 10: Pooled SVC effective-rank matching sensitivity. “Nearest” pairs each quantum candidate to its nearest classical rank with classical reuse; “one-to-one” uses minimum-cost assignment without reuse. Retained percentages use 64,800 possible quantum candidates as denominator. Δ is quantum minus classical OOD balanced accuracy.

Matcher	Caliper	Pairs (%)	Median ratio	95th pct. ratio	Median Δ	Q better
Nearest	1.10	27,928 (43.1)	1.040	1.093	−0.0054	29.4%
	1.25	48,708 (75.2)	1.082	1.226	−0.0056	29.7%
	1.50	60,647 (93.6)	1.112	1.390	−0.0061	28.9%
	2.00	64,408 (99.4)	1.123	1.542	−0.0070	27.8%
	none	64,800 (100)	1.124	1.578	−0.0071	27.6%
One-to-one	1.10	13,205 (20.4)	1.044	1.094	−0.0048	30.2%
	1.25	24,675 (38.1)	1.092	1.232	−0.0053	30.3%
	1.50	35,129 (54.2)	1.151	1.447	−0.0057	30.2%
	2.00	46,898 (72.4)	1.232	1.829	−0.0057	30.8%
	none	64,800 (100)	1.418	11.452	−0.0062	30.9%

At the primary 1.25 caliper, both matchers yield a negative group median in all eight groups.

The result is descriptive of the retained overlap, not a causal family estimate.

Supplementary Results: Geometry associations

Effective-rank association changes sign across regimes, whereas OOD kernel-target alignment is more consistently positive (Table 11). The source-dataset leave-one-out ordering correlations are 0.28 for EMBER, 0.22 for UNSW-NB15, and 0.40 for ToN-IoT when trained on the other two sources.

Supplementary Table 11: Geometry–OOD association and primary nearest-rank pairing for SVC. "Partial" controls kernel family and scale. Matching uses replacement and rank ratio ≤ 1.25; Δ is the median quantum-minus-classical OOD difference. All quantities are observational.

Scenario	ρ rank	Partial	ρ KTA	Matched Δ	Q better
EMBER $m1$	+0.32	+0.36	+0.30	−0.0094	33%
EMBER $m2$	+0.80	+0.82	+0.63	−0.0089	22%
UNSW-DoS (campaign)	+0.44	+0.41	+0.64	−0.0089	27%
UNSW-DoS (constructed)	−0.14	−0.08	+0.39	−0.0057	29%
UNSW-Recon (campaign)	+0.06	+0.06	+0.22	−0.0047	37%
UNSW-Recon (constructed)	−0.15	−0.08	+0.37	−0.0061	30%
ToN-IoT (campaign)	+0.25	+0.27	+0.46	−0.0060	24%
ToN-IoT (constructed)	+0.65	+0.63	+0.59	−0.0013	35%

Supplementary Results: Descriptive oracle comparison

The initial EMBER evaluation selected its representative using OOD test labels and fixed SVC at $C = 1$. Expanding the reference family nearly removes the apparent gain even before target-label-free selection is imposed.

Supplementary Table 12: Descriptive EMBER oracle results under fixed $C = 1$ (18 correlated settings). Mean and median effects are quantum minus classical balanced accuracy. These values are not deployable estimates or independent replicates.

Selection	Classifier	Classical reference	Q wins	Mean Δ	Median Δ
Best-by-OOD	SVC	linear+RBF	17/18	+0.0374	+0.0426
	SVC	extended	15/18	+0.0044	+0.0064
	GP	linear+RBF	15/18	+0.0283	+0.0339
	GP	extended	11/18	+0.0023	+0.0029
Best-by-ID	SVC	linear+RBF	18/18	+0.0775	+0.0794
	SVC	extended	18/18	+0.0149	+0.0152
	GP	linear+RBF	18/18	+0.0649	+0.0664
	GP	extended	18/18	+0.0192	+0.0166

Supplementary Results: Small-cluster interval sensitivity

Table 13 compares the primary pointwise conditional cluster-t interval with the prespecified BCa and leave-one-q-split diagnostics. The effective replication unit is five q-split clusters in every row. These intervals are not simultaneous and are not used as a familywise-error-controlled classification of the eight cases. Differences between interval constructions do not change the qualitative fixed-case description, and an interval containing zero is not interpreted as equivalence. With only five clusters, the BCa interval is retained as a fragile sensitivity rather than treated as additional confirmatory evidence.

Supplementary Table 13: Pointwise conditional uncertainty sensitivity for all primary equal-budget group–classifier cells. Δ is quantum minus classical OOD balanced accuracy. The cluster-t and 9,999-draw BCa columns are intervals over the same five q-split cluster means ($n = 5$); LOO is the range after leaving out one q-split cluster. All are conditional on the fixed benchmark pools and design, not intervals over a dataset population. They are not simultaneous intervals, and the BCa column is fragile at this effective sample size.

Scenario	Clf.	Δ	95% cluster-t	95% BCa	LOO range
EMBER $m1$	GPC	-0.0009	$[-0.0101, +0.0083]$	$[-0.0062, +0.0054]$	$[-0.0035, +0.0015]$
	SVC	-0.0058	$[-0.0340, +0.0223]$	$[-0.0226, +0.0110]$	$[-0.0120, +0.0014]$
EMBER $m2$	GPC	$+0.0027$	$[-0.0027, +0.0081]$	$[-0.0015, +0.0053]$	$[+0.0017, +0.0043]$
	SVC	-0.0064	$[-0.0137, +0.0009]$	$[-0.0107, -0.0018]$	$[-0.0081, -0.0049]$
UNSW-DoS (campaign)	GPC	-0.0057	$[-0.0120, +0.0005]$	$[-0.0084, +0.0004]$	$[-0.0078, -0.0046]$
	SVC	-0.0033	$[-0.0139, +0.0073]$	$[-0.0101, +0.0029]$	$[-0.0058, -0.0008]$
UNSW-DoS (constructed)	GPC	-0.0108	$[-0.0177, -0.0039]$	$[-0.0141, -0.0054]$	$[-0.0131, -0.0095]$
	SVC	-0.0044	$[-0.0119, +0.0031]$	$[-0.0087, +0.0005]$	$[-0.0067, -0.0026]$
UNSW-Recon (campaign)	GPC	$+0.0052$	$[-0.0019, +0.0123]$	$[-0.0001, +0.0089]$	$[+0.0038, +0.0072]$
	SVC	$+0.0050$	$[-0.0058, +0.0157]$	$[+0.0006, +0.0131]$	$[+0.0011, +0.0065]$
UNSW-Recon (constructed)	GPC	-0.0075	$[-0.0208, +0.0059]$	$[-0.0162, +0.0006]$	$[-0.0106, -0.0040]$
	SVC	-0.0089	$[-0.0146, -0.0031]$	$[-0.0123, -0.0053]$	$[-0.0102, -0.0076]$
ToN-IoT (campaign)	GPC	-0.0090	$[-0.0190, +0.0010]$	$[-0.0166, -0.0034]$	$[-0.0106, -0.0060]$
	SVC	-0.0148	$[-0.0268, -0.0029]$	$[-0.0218, -0.0068]$	$[-0.0180, -0.0129]$
ToN-IoT (constructed)	GPC	$+0.0109$	$[+0.0017, +0.0202]$	$[+0.0060, +0.0178]$	$[+0.0082, +0.0129]$
	SVC	-0.0011	$[-0.0049, +0.0028]$	$[-0.0033, +0.0013]$	$[-0.0019, -0.0003]$

Supplementary Table 14: Five q-split cluster means underlying each primary pointwise conditional interval. Values are quantum-minus-expected-classical OOD balanced-accuracy differences under P1$'$ and 60-candidate blocked matching. Lower-level model/master-seed and nested-size realizations are averaged within each displayed cluster; the five columns, not the individual pipeline realizations, are the effective replication units.

Scenario	Clf.	$q = 7$	$q = 42$	$q = 123$	$q = 999$	$q = 2024$
EMBER $m1$	SVC	-0.01393	$+0.01886$	-0.03483	-0.01476	$+0.01558$
	GPC	-0.00320	-0.00166	$+0.00135$	$+0.00965$	-0.01066
EMBER $m2$	SVC	-0.01221	-0.00951	-0.00064	$+0.00052$	-0.01010
	GPC	$+0.00351$	$+0.00663$	$+0.00625$	$+0.00098$	-0.00395
UNSW-DoS (campaign)	SVC	-0.00005	-0.01302	$+0.00666$	$+0.00135$	-0.01134
	GPC	-0.00595	-0.01046	-0.00637	-0.00859	$+0.00270$
UNSW-DoS (constructed)	SVC	-0.00466	-0.00689	-0.01185	$+0.00469$	-0.00340
	GPC	-0.01167	-0.01352	-0.01614	-0.00153	-0.01109
UNSW-Recon (campaign)	SVC	$+0.00246$	$+0.00188$	-0.00109	$+0.02025$	$+0.00127$
	GPC	$+0.00998$	$+0.00553$	-0.00281	$+0.01107$	$+0.00233$
UNSW-Recon (constructed)	SVC	-0.00424	-0.00367	-0.01388	-0.01128	-0.01122
	GPC	-0.01376	$+0.00122$	-0.00861	-0.02125	$+0.00511$
ToN-IoT (campaign)	SVC	-0.02155	-0.00193	-0.00700	-0.02105	-0.02260
	GPC	-0.02081	-0.00256	-0.00313	-0.01385	-0.00446
ToN-IoT (constructed)	SVC	$+0.00217$	-0.00268	-0.00324	$+0.00247$	-0.00403
	GPC	$+0.02184$	$+0.01485$	$+0.00872$	$+0.00624$	$+0.00304$

Supplementary Results: Quantum-map strata

Table 15 reports every fixed-case SVC effect after separating the 30 entangling-ZZ and 30 separable-product candidates. Each stratum is compared with the expected P1′ winner over 5,000 blocked draws of six complete extended-classical blocks, also 30 candidates. The full pool uses 60 candidates per family. Intervals are pointwise and conditional on five q-split clusters; their inclusion of zero is not an equivalence decision.

Supplementary Table 15: Circuit-stratified quantum-minus-classical OOD balanced-accuracy effects under train-CV and P1′. Brackets for scenario-groups are pointwise 95% conditional cluster-t intervals ($n = 5$ q-split clusters), not simultaneous intervals. The source-dataset-equal row gives the mean followed by the range of its three constituent source-dataset means, not a confidence interval.

Scenario	All maps ($B = 60$)	Entangling ZZ ($B = 30$)	Product maps ($B = 30$)
EMBER $m1$	−0.0057 [−0.0340, +0.0225]	−0.0063 [−0.0279, +0.0153]	+0.0036 [−0.0132, +0.0204]
EMBER $m2$	−0.0065 [−0.0138, +0.0008]	−0.0053 [−0.0130, +0.0024]	−0.0135 [−0.0166, −0.0104]
UNSW-DoS (campaign)	−0.0033 [−0.0139, +0.0073]	−0.0129 [−0.0184, −0.0073]	−0.0016 [−0.0116, +0.0083]
UNSW-DoS (constructed)	−0.0045 [−0.0119, +0.0030]	−0.0149 [−0.0299, +0.0001]	−0.0007 [−0.0063, +0.0049]
UNSW-Recon (campaign)	+0.0049 [−0.0058, +0.0157]	−0.0014 [−0.0070, +0.0041]	+0.0057 [−0.0023, +0.0137]
UNSW-Recon (constructed)	−0.0088 [−0.0145, −0.0032]	−0.0171 [−0.0286, −0.0056]	−0.0025 [−0.0066, +0.0015]
ToN-IoT (campaign)	−0.0149 [−0.0270, −0.0028]	−0.0196 [−0.0303, −0.0089]	−0.0225 [−0.0347, −0.0102]
ToN-IoT (constructed)	−0.0011 [−0.0050, +0.0028]	−0.0013 [−0.0047, +0.0021]	−0.0009 [−0.0036, +0.0018]
Source-dataset-equal	−0.0057 [−0.0080, −0.0029]	−0.0093 [−0.0116, −0.0058]	−0.0055 [−0.0117, +0.0002]

For P1′ selection, product maps account for 594/1,080 SVC winners and 348/1,080 GPC winners. Under same-test OOD selection, the corresponding counts are 650/1,080 and 528/1,080. Complete counts by individual map are shown in Table 16 and provided in `quantum_winner_composition.csv`.

Supplementary Table 16: Quantum-map composition of the within-family winner. Each model–selector column contains 1,080 fixed security runs. Counts and percentages describe which configuration maximizes the specified selection metric within the quantum family; they do not compare its OOD accuracy with the classical family. P1′ uses disjoint ID validation, whereas the OOD oracle reuses the shifted-test labels.

Map	Stratum	P1′ SVC	P1′ GPC	OOD-oracle SVC	OOD-oracle GPC
ZZ, 1 rep.	entangling ZZ	290 (26.9%)	353 (32.7%)	282 (26.1%)	279 (25.8%)
ZZ, 2 reps.	entangling ZZ	196 (18.1%)	379 (35.1%)	148 (13.7%)	273 (25.3%)
Pauli-X/Z, 1 rep.	separable product	315 (29.2%)	171 (15.8%)	353 (32.7%)	246 (22.8%)
Z, 2 reps.	separable product	279 (25.8%)	177 (16.4%)	297 (27.5%)	282 (26.1%)

Supplementary Results: Within-v4 factorial

The q1000 factorial holds the v4 data generation, split roles, kernel inventory, and preprocessing fixed while crossing regularization, selector, classical reference, and budget mode. Table 17 reports all 16 three-source-dataset-equal endpoints. The fixed-C, same-test-oracle, customary-reference, native-budget corner is +0.0131; the train-CV, ID-validation, extended-reference, equal-count corner is −0.0047. These endpoints are fixed-case summaries, not population effects.

Table 18 summarizes paired axis changes and pairwise differences-in-differences. Each axis row averages its paired change over the other eight factorial cells; each interaction row averages over the four cells of the remaining two axes. The relatively large regularization-by-reference interaction shows why the axis averages are not interpreted as independent, additive, or causal contributions.

Supplementary Results: Network-field ablation

The frozen q1000 sensitivity removes port-, protocol-, and service-related fields before fitting the shared train-only embedding. The ablation changes absolute task difficulty, especially for the

Supplementary Table 17: Within-v4 q1000 SVC factorial. Entries are three-source-dataset-equal quantum-minus-classical OOD balanced-accuracy effects. "Native" uses the complete family pools; "equal count" block-samples the larger family. Positive values favour the fidelity-kernel family.

Regularization and selector	Customary native	Customary equal	Extended native	Extended equal
$C = 1$, OOD oracle	+0.0131	+0.0094	−0.0010	+0.0021
$C = 1$, ID validation	+0.0118	+0.0118	−0.0079	−0.0068
Train-CV, OOD oracle	+0.0018	−0.0012	−0.0066	−0.0040
Train-CV, ID validation	−0.0074	−0.0084	−0.0040	−0.0047

Supplementary Table 18: Descriptive factorial axis changes and pairwise interactions. Changes are level-two minus level-one quantum-minus-classical effects, so a negative value is a shift towards the classical family. Interactions are differences-in-differences. No population inference, causal attribution, or familywise decision rule is attached to these finite-design summaries.

Finite-design summary	Mean change
Regularization: $C = 1 \to$ train-CV	−0.0084
Selection: OOD oracle → ID validation	−0.0037
Reference: customary → extended	−0.0079
Budget: native → equal count	−0.0002
Interaction: reference × budget mode	+0.0034
Interaction: regularization × budget mode	−0.0007
Interaction: regularization × reference	+0.0139
Interaction: regularization × selection	+0.0000
Interaction: selection × budget mode	+0.0001
Interaction: selection × reference	+0.0004

ToN-IoT campaign shift, and can affect the two kernel families differently. Table 19 therefore reports original, ablated, and paired-change effects separately.

Supplementary Table 19: Port/protocol/service-field ablation under train-CV, P1′, and blocked 60-candidate matching (SVC). Scenario-group brackets are pointwise 95% conditional cluster-t intervals over five q-split clusters and are not simultaneous. The final-row bracket is the range of the UNSW-NB15 and ToN-IoT source-dataset changes, not a confidence interval.

Scenario	Original effect	Ablated effect	Ablated minus original
UNSW-DoS (campaign)	−0.0021 [−0.0208, +0.0167]	−0.0108 [−0.0231, +0.0016]	−0.0087 [−0.0234, +0.0061]
UNSW-DoS (constructed)	+0.0014 [−0.0132, +0.0161]	+0.0008 [−0.0264, +0.0280]	−0.0006 [−0.0234, +0.0221]
UNSW-Recon (campaign)	−0.0028 [−0.0275, +0.0220]	+0.0072 [−0.0091, +0.0234]	+0.0099 [−0.0268, +0.0466]
UNSW-Recon (constructed)	−0.0138 [−0.0284, +0.0007]	−0.0074 [−0.0289, +0.0141]	+0.0065 [−0.0186, +0.0315]
ToN-IoT (campaign)	−0.0007 [−0.0111, +0.0097]	−0.0801 [−0.1717, +0.0114]	−0.0794 [−0.1753, +0.0165]
ToN-IoT (constructed)	−0.0029 [−0.0095, +0.0037]	−0.0129 [−0.0516, +0.0259]	−0.0100 [−0.0459, +0.0259]
Two-source-dataset-equal	−0.0031	−0.0245	−0.0215 [−0.0447, +0.0018]

Supplementary Results: Circuit resources

Supplementary Table 20: Complete logical circuit audit. Counts follow Qiskit 2.3.1 decomposition to `rz`, `sx`, `x`, and `cx` at optimization level 0 without device routing. "Map" is one feature-map state preparation; "fidelity" is its compute–uncompute template for one pair of inputs. One-qubit counts exclude barriers and measurements. These compiler-conditional logical counts are not device costs.

Map	Qubits	Map depth	Map 1q	Map CX	Fidelity depth	Fidelity 1q	Fidelity CX
ZZ, 1 rep.	4	19	22	12	38	44	24
	6	31	39	30	62	78	60
	8	43	60	56	86	120	112
	10	55	85	90	110	170	180
	12	67	114	132	134	228	264
ZZ, 2 reps.	4	35	44	24	70	88	48
	6	53	78	60	106	156	120
	8	71	120	112	142	240	224
	10	89	170	180	178	340	360
	12	107	228	264	214	456	528
Pauli-X/Z, 1 rep.	4	11	44	0	22	88	0
	6	11	66	0	22	132	0
	8	11	88	0	22	176	0
	10	11	110	0	22	220	0
	12	11	132	0	22	264	0
Z, 2 reps.	4	8	32	0	16	64	0
	6	8	48	0	16	96	0
	8	8	64	0	16	128	0
	10	8	80	0	16	160	0
	12	8	96	0	16	192	0

Supplementary Results: External fixed tasks

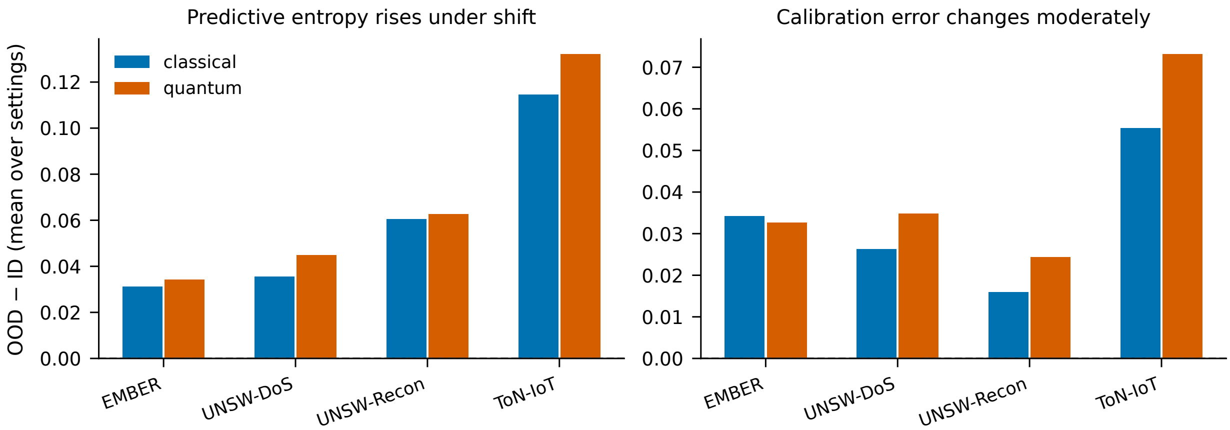
All 30 prospectively specified task–size–seed units passed the acquisition, split-integrity, nesting, and leakage audits. They contain 37,800 split-level results. Under SVC, size-equal controlled effects for College Scorecard, diabetes readmission, and ACS income are -0.0160, -0.0066, and -0.0131; their protocol contractions are $+0.0393$, $+0.0105$, and $+0.0126$. The task-equal controlled effect is -0.0119 (95% conditional seed-cluster interval $[-0.0205, -0.0033]$, $n = 5$ seeds per task), and contraction is $+0.0208$ ($[+0.0113, +0.0307]$). Under GPC, the task-equal controlled effect is $+0.0077$ ($[-0.0015, +0.0183]$) and contraction is $+0.0324$ ($[+0.0223, +0.0427]$). These fixed-task results corroborate protocol sensitivity, not a universal family ordering.

Supplementary Results: Predictive uncertainty

Predictive entropy rises from ID to OOD for both honestly selected families in every scenario-group. Expected calibration error changes moderately and comparably (Figure 2). Together with the controlled accuracy results, these diagnostics provide no consistent evidence in the fixed cases of a calibration benefit unique to the selected fidelity kernels. Entropy increase alone is not evidence of good calibration.

Supplementary Results: Repeated finite-shot sensitivity

Figure 3 displays all 2,880 finite-shot evaluations: eight fixed exact Gram matrices, four shot counts, 30 measurement replicates, and three PSD-handling conditions. The bands make the conditional Monte Carlo spread visible rather than treating diversity across data configurations as measurement replication. The independent-square heuristic and coherent train-based Nyström extension often



Supplementary Figure 2: Mean ID-to-OOD change in predictive entropy (left) and 15-bin expected calibration error (right) for the P1′-selected configuration of each family. Bars aggregate fixed scenario-groups by source dataset and are descriptive; no population interval is implied.

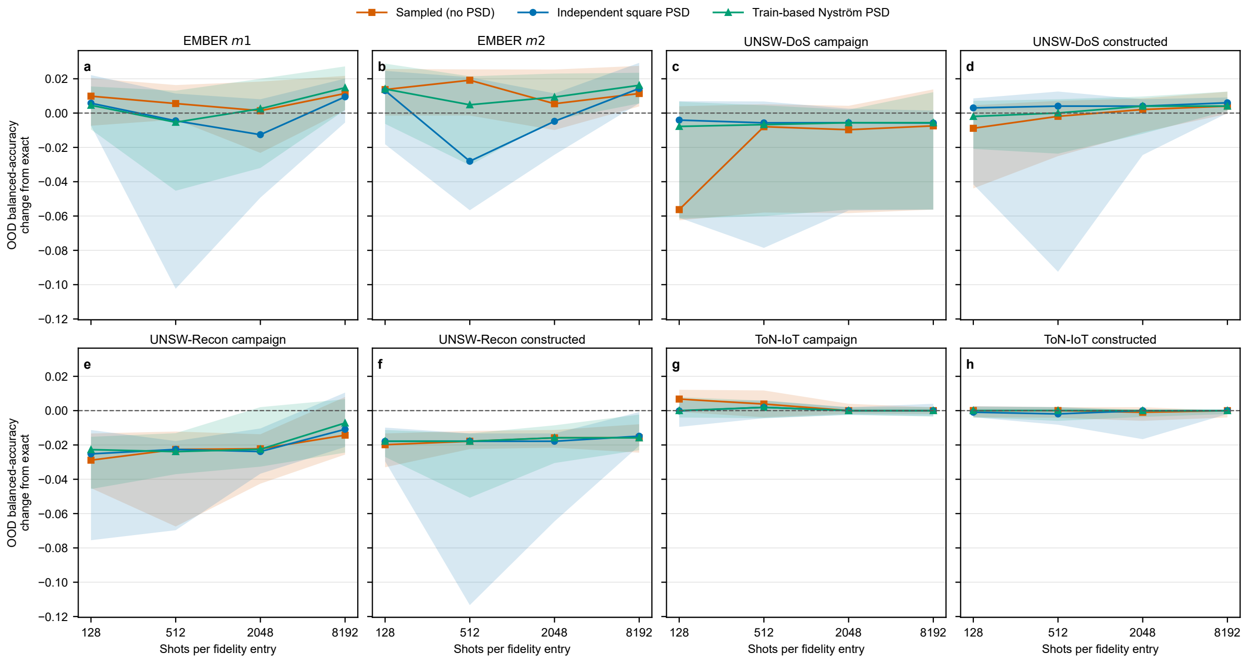
narrow or shift the distribution but do not impose a uniform direction. Exact-statevector OOD accuracy is approached unevenly because each sampled training matrix reselects a discrete SVC C.

Across the eight group medians, the perturbed/exact effective-rank ratio declines from 1.93 at 128 shots to 1.13 at 8,192 shots. OOD-KTA under the Nyström extension is estimator-dependent because its OOD square is derived from OOD–train features rather than independently sampled; it is not treated as a mechanism or compared directly with the other KTA estimators. Retained rank, eigenvalue tolerance, blockwise Frobenius changes, negative-spectrum fractions, selected C, and every replicate endpoint are provided in `results/v6/shots_mc/`.

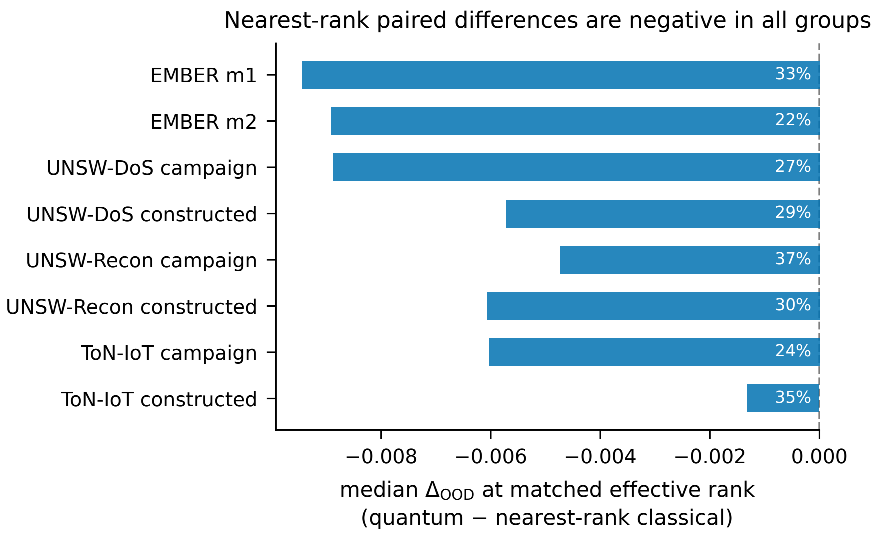
Supplementary Table 21: Projected sampling resources for the repeated finite-shot sensitivity. One case–replicate contains 1,624,250 distinct fidelity parameterizations. "All eight" multiplies one shot level by eight fixed cases and 30 measurement replicates; "entangling four" retains only the four selected ZZ cases. The other four selected configurations are separable-product maps whose direct classical factorization can bypass circuit sampling. The three PSD conditions reuse these measurements. Counts exclude device routing, calibration, mitigation, queueing, and repeated-job overhead; no hardware execution was performed.

Shots/fidelity	Shots/case–replicate	Estimation instances/all eight	Shots/all eight	Shots/entangling four
128	207,904,000	389,820,000	49,896,960,000	24,948,480,000
512	831,616,000	389,820,000	199,587,840,000	99,793,920,000
2,048	3,326,464,000	389,820,000	798,351,360,000	399,175,680,000
8,192	13,305,856,000	389,820,000	3,193,405,440,000	1,596,702,720,000
All four	17,671,840,000	1,559,280,000	4,241,241,600,000	2,120,620,800,000

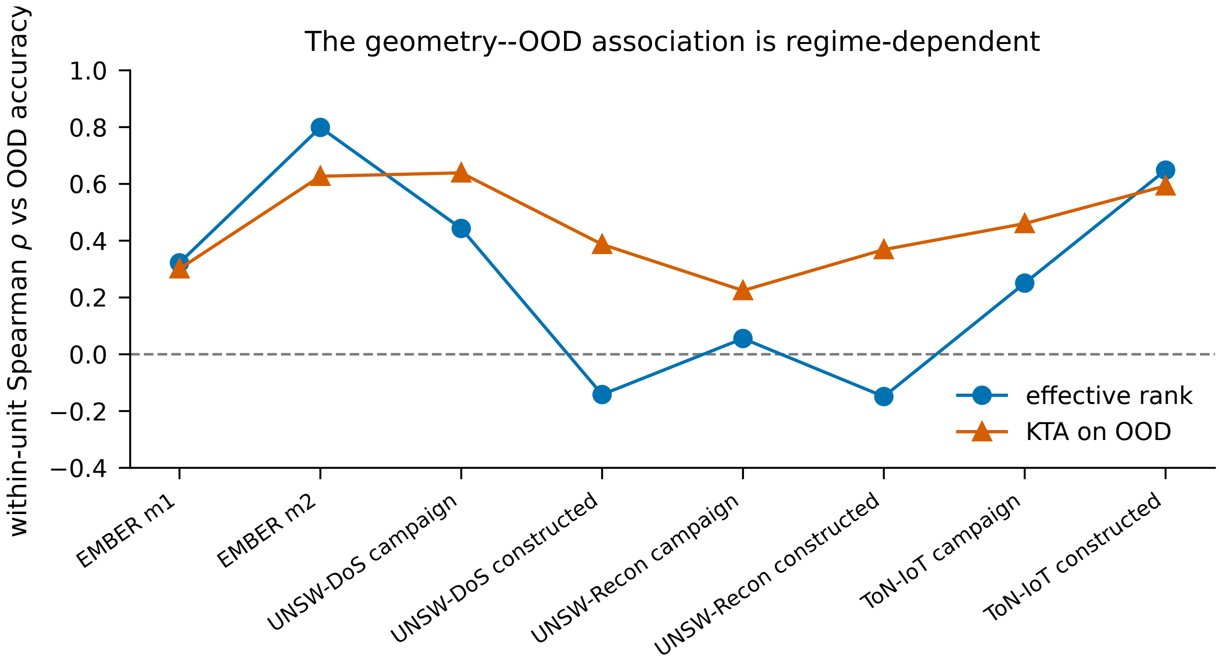
Supplementary Results: Geometry displays moved from the main text



Supplementary Figure 3: Repeated finite-shot OOD change from the exact fixed-configuration endpoint. Panels show one fixed scenario-group Gram matrix. Lines are medians over 30 independently seeded measurement replicates and shaded regions are central 95% Monte Carlo intervals ($n = 30$) conditional on that matrix. Orange squares use the sampled indefinite matrix; blue circles use independent PSD projections of train and OOD squares while leaving rectangular blocks unchanged; green triangles use the coherent train-eigenspace Nyström extension. The model includes fidelity sampling only, not device noise.



Supplementary Figure 4: Nearest-rank pairing yields negative median quantum-minus-classical differences in all eight fixed scenario-groups. Each quantum configuration is paired, with replacement, to the nearest-effective-rank classical configuration within run and dimension, subject to rank ratio ≤ 1.25. Annotations give the fraction of retained pairs in which the quantum member is more accurate. The pairing is observational and does not imply a causal family effect.



Supplementary Figure 5: The geometry–OOD association is regime-dependent. Points are median within-unit Spearman correlations between each descriptor and OOD balanced accuracy for SVC. Effective rank is strong in some regimes and weak or negative in others; OOD alignment is more consistently positive but reuses the target labels. These descriptive associations do not identify a causal mechanism.

Supplementary Table 22: Zero-label certificate across the frozen nested reference breadths. The endpoint column is the median across the 16 cell-level medians; the range is across those cell medians. Final columns count cells whose median upper endpoint is at or below each materiality threshold. Every 30- and 60-candidate cell uses 5,000 whole-block orderings; breadth 115 contains the complete family in every ordering.

B	Blocks	Median $U_B^{(0)}$	Cell range	$U \leq .005$	$U \leq .010$	$U \leq .020$
30	6	.042	.004–.094	2	3	3
60	12	.034	.002–.088	2	4	5
115	23	.034	.002–.088	2	4	5

Supplementary Results: Machine-readable result map

The v0.9 retrospective artifacts and manuscript gates are reproduced by

```
python scripts/reproduce_v9.py --stage analysis
python scripts/reproduce_v9.py --stage report
python scripts/reproduce_v9.py --stage audit
python scripts/reproduce_v10.py --stage all
python scripts/reproduce_v11.py --stage all
```

The exact-prediction export is separately available as `python scripts/reproduce_v9.py --stage export`; it skips complete, integrity-gated case artifacts unless forced. Target prediction matrices, separated audit labels, sharp envelopes, evidence-frontier curves, and finite-shot certificate outputs are under `results/v9/partial_identification/`. Circuit-aware strata and the within-v4 factorial remain under `results/v8/`; prior inference, budget, rank, and finite-shot inputs remain under `results/v6/`. Frozen earlier artifacts are unchanged. The development repository is `https://github.com/roberto-fernandez-barrios/kernel_shift_framework`, and the immutable v1.1.1 archive is `https://doi.org/10.5281/zenodo.21751137`; the all-version concept DOI is `https://doi.org/10.5281/zenodo.19147649`. The exact v0.8.0 predecessor is identified by `https://doi.org/10.5281/zenodo.21717074`. The v1.0 prospective inputs and outputs are under `results/v10/gate2_prospective/`; the frozen specification, acquisition audits, prediction locks, opaque label hashes, one-time opening record, all policy draws, technical-unavailability records, and final outcome are included in the v1.1.1 archive together with the bounded-loss tests, consolidation specification, manuscript sources, and final PDFs.